

Calculations:

(i): Current Threshold (power limit)

From gn:

$$P_{\max} = 5 \text{ KW} = 5000 \text{ W}$$

$$\text{Max voltage} = 400 \text{ V}$$

$$I = \frac{P}{V}$$

$$I = \frac{5000}{400}$$

~~$$I = 12.5 \text{ A}$$~~

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For our project: Our sensor mapping:

$$0 \rightarrow 50 \text{ A} \Rightarrow 0 \rightarrow 5 \text{ V}$$

$$\frac{5 \text{ V}}{50 \text{ A}} = 0.1 \frac{\text{V}}{\text{A}}$$

$$V_{\text{current}} = 12.5 \times 0.1 \\ = 1.25 \text{ V}$$

Comparator
reference for
Current

i) Brake Pressure Threshold:-

* Max brake pressure = 30 bar

* Sensor mapping: $0 \rightarrow 50 \text{ bar} \rightarrow 0-5 \text{ V}$

$$\frac{5 \text{ V}}{50 \text{ bar}} \Rightarrow 0.1 \frac{\text{V}}{\text{bar}}$$

$$V_{\text{brake}} = 30 \times 0.1 = 3.0 \text{ V}$$

Comparator reference for brake = 3V

(iii): RC Delay (500ms condition)

$$V(t) = V_{cc}(1 - e^{-t/R})$$

Let's take component values:

$$R = 100 \text{ k}\Omega$$

$$C = 4.7 \mu\text{F}$$

$$t \approx RC$$

Time Constant Calculation:

$$t = RC$$

$$t = 100\,000 \times 4.7 \times 10^{-6}$$

$$t = 0.47 \text{ s}$$

$$\boxed{t = 470 \text{ ms}}$$

Designed delay = 470ms which is close to required 500ms

(iv): Boolean Logic Expression:

B = Brake high (Pressure > 30 bar)

C = Current (> 5kW equivalent)

T = Time Delay (> 500ms)

The logical condition we used for comparator is

$$\text{BSPD} = B \cdot C \cdot T$$

In simple terms both brake and current must be high simultaneously and should be more than 500ms then shut down circuit opens (SDC)

(V) Relay Driver design:

BSPD output drives a relay to break down the shutdown circuit.

Soln: $I_c = 70 \text{ mA} \Rightarrow$ Relay coil current.

Transistor gain: $\beta = 100$

Base current calculation:

$$I_B = \frac{I_c}{\beta}$$

$$I_B = \frac{70 \text{ mA}}{100} = 0.7 \text{ mA}$$

For saturation condition, let's have higher base current:

$$I_B \approx 1 \text{ mA}$$

Base Resistor Calculations:

$$R_B = \frac{V_{in} - V_{BE}}{I_B}$$

$$R_B = \frac{5 - 0.7}{1 \text{ mA}}$$

$$R_B \approx 4.3 \text{ k}\Omega$$

So we choose $R_B \approx 4.7 \text{ k}\Omega$

For practical application.

Final Working:

Condition

Output:

Fault Detected

Transistor ON $\rightarrow$ Relay ON $\rightarrow$ SDL opened.

No fault

Transistor off $\rightarrow$ Relay off $\rightarrow$ Normal operation.